

$$(25) \int \tan^5 x \cdot \sec^6 x dx$$

$$\tan^2 x = u \Rightarrow \tan x \sec^2 x dx = \frac{du}{2} \quad \sec^2 x = 1 + u$$

$$\begin{aligned} \int \tan^5 x \sec^6 x dx &= \int \tan^4 x \cdot \sec^4 x \tan x \sec^2 x dx = \int u^2 (1+u)^2 du \\ &= \int (u^6 + 2u^4 + u^2) du = \frac{1}{7} u^7 + \frac{2}{5} u^5 + \frac{1}{3} u^3 + C \\ &= \frac{1}{7} \tan^{14} x + \frac{2}{5} \tan^{10} x + \frac{1}{3} \tan^6 x + C // \end{aligned}$$

$$(26) \int \sqrt{\frac{a+x}{a-x}} dx$$

$$x = a \cos 2t = a(\cos^2 t - \sin^2 t)$$

$$dx = -2a \sin 2t dt$$

$$a+x = a(1 + \cos^2 t - \sin^2 t) = 2a \cos^2 t$$

$$a-x = a(1 - \cos^2 t + \sin^2 t) = 2a \sin^2 t$$

$$\int \sqrt{\frac{a+x}{a-x}} dx = \int \sqrt{\frac{2a \cos^2 t}{2a \sin^2 t}} (-2a \sin 2t) dt = -2a \int \frac{\cos t}{\sin t} \cdot 2 \sin t \cos t dt$$

$$= -4a \int \cos^2 t dt = -4a \int \frac{1 + \cos 2t}{2} dt = -2a \left(t + \frac{\sin 2t}{2} \right) + C$$

$$= -2at - a \sin 2t + C = -a \arccos \frac{x}{a} - a \sqrt{1 - \left(\frac{x}{a}\right)^2} + C$$

$$= -a \arccos \frac{x}{a} - \sqrt{a^2 - x^2} + C$$

$$(27) \int \tan^2(e^x) e^x dx$$

$$e^x = u \quad e^x dx = du$$

$$\int \tan^2(e^x) e^x dx = \int \tan^2 u du = \int (\sec^2 u - 1) du = \tan u - u + C$$

$$= \tan(e^x) - e^x + C //$$

$$28) \int \frac{2^x}{4^{x+1}} dx$$

$$2^x = \tan t \Rightarrow 2^x \ln 2 dx = \sec^2 t dt \Rightarrow 2^x dx = \frac{1}{\ln 2} \sec^2 t dt$$

$$\int \frac{2^x}{4^{x+1}} dx = \frac{1}{\ln 2} \int \frac{\sec^2 t dt}{\sec^2 t} = \frac{1}{\ln 2} t + C = \frac{1}{\ln 2} \arctan 2^x + C$$

II. 461:

$$2^x = u \Rightarrow 2^x \ln 2 dx = du \Rightarrow 2^x dx = \frac{1}{\ln 2} du$$

$$\int \frac{2^x}{4^{x+1}} dx = \frac{1}{\ln 2} \int \frac{du}{u^2+1} = \frac{1}{\ln 2} \arctan u = \frac{1}{\ln 2} \arctan 2^x + C$$

$$29) \int \frac{dx}{e^x+1} = \int \frac{dx}{e^x(1+e^{-x})}$$

$$1+e^{-x} = u \Rightarrow e^{-x} dx = -du$$

$$\int \frac{dx}{e^x+1} = - \int \frac{du}{u} = -\ln|u| + C = -\ln|1+e^{-x}| + C$$

$$30) I = \int \frac{\sqrt{9-x^2}}{x} dx$$



$$x = 3 \sin t \Rightarrow dx = 3 \cos t dt$$

$$I = \int \frac{\sqrt{9-x^2}}{x} dx = \int \frac{3 \cos t}{3 \sin t} 3 \cos t dt = 3 \int \frac{\cos^2 t}{\sin t} dt = 3 \int \csc t - 3 \int \sin t dt$$

$$= 3 \int \frac{\csc t (\csc t - \cot t)}{(\csc t - \cot t)} dt - 3 \int \sin t dt$$

$$= 3 \ln |\csc u - \cot u| + 3 \cos u + C = 3 \ln \left| \frac{3}{x} - \frac{\sqrt{9-x^2}}{x} \right| + \sqrt{9-x^2} + C$$

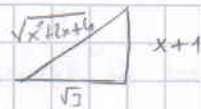
II. 461: $3 \int \frac{\cos^2 t}{\sin t} dt = 3 \int \frac{\cos^2 t \sin t dt}{1 - \cos^2 t} = 3 \int \frac{u^2 du}{u^2 - 1} = 3 \int \left(1 + \frac{1}{u^2 - 1} \right) du$

$$= 3 \int du + \frac{3}{2} \int \frac{du}{u-1} - \frac{3}{2} \int \frac{du}{u+1} = 3u + \frac{3}{2} \ln \left| \frac{u-1}{u+1} \right| + C$$

$$= 3 \cos t + \frac{3}{2} \ln \left| \frac{\cos t - 1}{\cos t + 1} \right| + C = \dots$$

$$(31) I = \int \sqrt{x^2 + 2x + 4} dx$$

$$x+1 = \sqrt{3} \tan t \Rightarrow dx = \sqrt{3} \sec^2 t dt$$



$$I = \int \sqrt{x^2 + 2x + 4} dx = \int \sqrt{(x+1)^2 + 3} dt = \int \sqrt{3} \sec t \cdot \sqrt{3} \sec^2 t dt$$

$$= 3 \int \sec^3 t dt \stackrel{\text{örnek 10}}{=} \frac{3}{2} \left[\sec t \tan t + \ln |\sec t + \tan t| \right] + c$$

$$= \frac{3}{2} \left[\frac{\sqrt{x^2 + 2x + 4}}{\sqrt{3}} \cdot \frac{x+1}{\sqrt{3}} + \ln \left| \frac{\sqrt{x^2 + 2x + 4}}{\sqrt{3}} + \frac{x+1}{\sqrt{3}} \right| \right] + c$$

$$= \frac{1}{2} \sqrt{x^2 + 2x + 4} (x+1) + \frac{3}{2} \ln \left| \frac{\sqrt{x^2 + 2x + 4} + x + 1}{\sqrt{3}} \right| + c$$

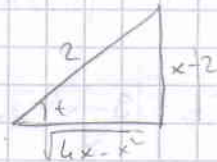
$$(32) \int \frac{dx}{(4x - x^2)^{3/2}}$$

$$x-2 = 2 \sin t \Rightarrow dx = 2 \cos t dt$$

$$4x - x^2 = 4 - (x-2)^2$$

$$\int \frac{dx}{(4x - x^2)^{3/2}} = \int \frac{2 \cos t dt}{2 \cos^3 t dt} = \int \sec^2 t dt = 4 - 4 \sin^2 t = 4 \cos^2 t$$

$$= \tan t + c = \frac{x-2}{\sqrt{4x-x^2}} + c$$



$$(33) \int \frac{dx}{x \sqrt{x^2 - 3x + 2}} = \int \frac{dx}{x^2 \sqrt{1 - \frac{3}{x} + \frac{2}{x^2}}} = - \int \frac{du}{\sqrt{1 - 3u + 2u^2}} = - \frac{1}{\sqrt{2}} \int \frac{du}{\sqrt{u^2 - \frac{3}{2}u + \frac{1}{2}}}$$

$$u = \frac{1}{x}$$

$$du = \frac{dx}{x^2}$$

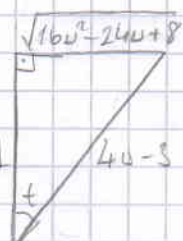
$$= - \frac{1}{\sqrt{2}} \int \frac{du}{\sqrt{(u - \frac{3}{4})^2 - \frac{1}{16}}} = - \frac{1}{\sqrt{2}} \int \frac{\frac{1}{4} \sec t \tan t dt}{\frac{1}{4} \tan t}$$

$$u - \frac{3}{4} = \frac{1}{4} \sec t$$

$$= - \frac{1}{\sqrt{2}} \int \sec t dt = - \frac{1}{\sqrt{2}} \ln |\sec t + \tan t| + c$$

$$\frac{du}{4} = \frac{1}{4} \sec t \tan t$$

$$= - \frac{1}{\sqrt{2}} \ln \left| 4u - 3 + \frac{\sqrt{16u^2 - 24u + 8}}{4u - 3} \right| + c$$



$$= - \frac{1}{\sqrt{2}} \ln \left| \frac{4}{x} - 3 + \frac{\sqrt{16/x^2 - 24/x + 8}}{4/x - 3} \right| + c$$

$$= - \frac{1}{\sqrt{2}} \ln \left| \frac{4}{x} - 3 + \frac{\sqrt{8x^2 - 24x + 16}}{4 - 3x} \right| + c$$

II. Teil:

$$x - \frac{3}{2} = \frac{1}{2} \sec t \Rightarrow dx = \frac{1}{2} \sec t \tan t dt$$

$$\begin{aligned} \int \frac{dx}{x\sqrt{x^2-3x+2}} &= \int \frac{dx}{x\sqrt{(x-\frac{3}{2})^2 - \frac{1}{4}}} \\ &= \int \frac{\frac{1}{2} \sec t \tan t dt}{(\frac{3}{2} + \frac{1}{2} \sec t) \sqrt{\frac{1}{4} \sec^2 t - \frac{1}{4}}} = \int \frac{\frac{1}{2} \sec t \tan t dt}{(\frac{3}{2} + \frac{1}{2} \sec t) \frac{1}{2} \tan t} \\ &= 2 \int \frac{dt}{3 \cos t + 1} = 2 \int \frac{2 du / (1+u^2)}{3 - 3u^2 + 1 + u^2} = 4 \int \frac{du}{8 - 3u^2 + 1 + u^2} \\ &= 2 \int \frac{du}{2 - u^2} = \frac{-2}{2\sqrt{2}} \left\{ \int \frac{du}{u+\sqrt{2}} - \int \frac{du}{u-\sqrt{2}} \right\} \\ &= \frac{-1}{\sqrt{2}} (\ln|u+\sqrt{2}| - \ln|u-\sqrt{2}|) + c \quad \text{--- (1)} \end{aligned}$$



$$\tan \frac{t}{2} = u$$

$$\sin t = \frac{2u}{1+u^2}$$

$$\cos t = \frac{1-u^2}{1+u^2}$$

$$dt = \frac{2 du}{1+u^2}$$

$$u = \tan \frac{t}{2} = \frac{\sin t/2}{\cos t/2} = \frac{\sqrt{\frac{1-\cos t}{2}}}{\sqrt{\frac{1+\cos t}{2}}} = \frac{\sqrt{1-\cos t}}{\sqrt{1+\cos t}} = \frac{\sqrt{1 - \frac{2x-3}{2}}}{\sqrt{1 + \frac{2x-3}{2}}} = \frac{\sqrt{2x-6}}{\sqrt{2x-2}} = \sqrt{\frac{x-3}{x-1}} \quad \text{--- (2)}$$

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$$I = \frac{-1}{\sqrt{2}} \left[\ln \left| \sqrt{\frac{x-2}{x-1}} + \sqrt{2} \right| - \ln \left| \sqrt{\frac{x-2}{x-1}} - \sqrt{2} \right| \right] + c$$

$$= \frac{-1}{\sqrt{2}} \left[\ln \left| \frac{\sqrt{x-2} + \sqrt{2x-2}}{\sqrt{x-1}} \right| - \ln \left| \frac{\sqrt{x-2} - \sqrt{2x-2}}{\sqrt{x-1}} \right| \right] + c$$

$$= \frac{-1}{\sqrt{2}} \ln \left| \frac{\sqrt{x-2} + \sqrt{2x-2}}{\sqrt{x-2} - \sqrt{2x-2}} \right| + c = \frac{-1}{\sqrt{2}} \ln \left| \frac{(\sqrt{x-2} + \sqrt{2x-2})^2}{x-2 - (2x-2)} \right| + c$$

$$= \frac{-1}{\sqrt{2}} \ln \left| \frac{x-2 + 2x-2 + 2\sqrt{2x^2-6x+4}}{-x} \right| + c$$

$$= \frac{-1}{\sqrt{2}} \ln \left| \frac{\frac{4}{x} - 3 + \sqrt{8x^2-24x+16}}{-x} \right| + c$$

$$(3u) \int \frac{dx}{x\sqrt{x^2-x+2}} = \int \frac{dx}{x^2\sqrt{1-\frac{1}{x}+2/x^2}} = - \int \frac{du}{\sqrt{1-u+2u^2}} = -\frac{1}{\sqrt{2}} \int \frac{du}{\sqrt{u^2-\frac{u}{2}+\frac{1}{2}}}$$

$$u = \frac{1}{x} \\ -du = \frac{dx}{x^2}$$

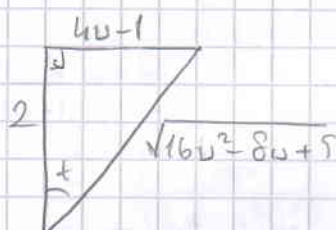
$$= -\frac{1}{\sqrt{2}} \int \frac{du}{\sqrt{(u-\frac{1}{4})^2 + \frac{1}{4}}} = -\frac{1}{\sqrt{2}} \int \frac{\frac{1}{2} \sec^2 t dt}{\frac{1}{2} \sec t} = -\frac{1}{\sqrt{2}} \int \sec t dt$$

$$u - \frac{1}{4} = \frac{1}{2} \tan t$$

$$= -\frac{1}{\sqrt{2}} \ln |\sec u + \tan u| + C = -\frac{1}{\sqrt{2}} \ln \left| \frac{\sqrt{16u^2-8u+5}}{2} + \frac{4u-1}{2} \right| + C$$

$$du = \frac{1}{2} \sec^2 t dt$$

$$= -\frac{1}{\sqrt{2}} \ln \left| \sqrt{\frac{16}{x^2} - \frac{8}{x} + 5} + \frac{4}{x} - 1 \right| - \frac{1}{\sqrt{2}} \ln 2 + C$$



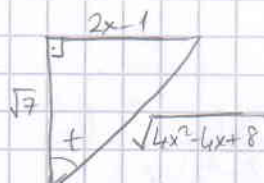
$$= -\frac{1}{\sqrt{2}} \ln |\sqrt{5x^2-8x+16} + 4-x| - \frac{1}{\sqrt{2}} \ln |x| + C$$

II. Sol:

$$\int \frac{dx}{x\sqrt{x^2-x+2}} = \int \frac{dx}{x\sqrt{(x-\frac{1}{2})^2 + \frac{7}{4}}} = \int \frac{\frac{\sqrt{7}}{2} \sec^2 t dt}{(\frac{1}{2} + \frac{\sqrt{7}}{2} \tan t) \frac{\sqrt{7}}{2} \sec t} = 2 \int \frac{dt}{\cos t + \sqrt{7} \sin t}$$

$$\frac{x-1}{2} = \frac{\sqrt{7}}{2} \tan t$$

$$\frac{dx}{2} = \frac{\sqrt{7}}{2} \sec^2 t dt$$



$$= 2 \int \frac{\frac{2du}{1+u^2}}{\frac{1-u^2}{1+u^2} + \frac{2\sqrt{7}u}{1+u^2}} = 4 \int \frac{du}{1-2\sqrt{7}u-u^2} = 4 \int \frac{du}{8-(u-\sqrt{7})^2}$$

$$= \frac{4}{8} \int \frac{du}{1 - \left(\frac{u-\sqrt{7}}{2\sqrt{2}}\right)^2} = \frac{1}{2} \int \frac{2\sqrt{2} dv}{1-v^2}$$

$$v = \frac{u-\sqrt{7}}{2\sqrt{2}}$$

$$du = 2\sqrt{2} dv$$

$$\tan \frac{t}{2} = u$$

$$= -\sqrt{2} \int \frac{dv}{v^2-1} = -\frac{\sqrt{2}}{2} \left\{ \int \frac{dv}{v-1} - \int \frac{dv}{v+1} \right\}$$

$$dt = \frac{2du}{1+u^2}$$

$$= -\frac{\sqrt{2}}{2} \left\{ \ln |v-1| - \ln |v+1| \right\} + C = -\frac{\sqrt{2}}{2} \left\{ \ln \left| \frac{u-\sqrt{7}}{2\sqrt{2}} - 1 \right| - \ln \left| \frac{u-\sqrt{7}}{2\sqrt{2}} + 1 \right| \right\} + C$$

$$\sin t = \frac{2u}{1+u^2}$$

$$I = -\frac{\sqrt{2}}{2} \left\{ \ln |u-\sqrt{7}-2\sqrt{2}| - \ln |u-\sqrt{7}+2\sqrt{2}| \right\} + \frac{\sqrt{2}}{2} \ln 2\sqrt{2} + C$$

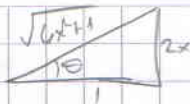
$$\cos t = \frac{1-u^2}{1+u^2}$$

$$u = \tan \frac{t}{2} = \frac{1-\cos t}{1+\cos t} = \frac{1-\sqrt{7}/\sqrt{4x^2-4x+8}}{1+\sqrt{7}/\sqrt{4x^2-4x+8}} = \frac{\sqrt{4x^2-4x+8}-\sqrt{7}}{2x-1}$$

$$I = -\frac{\sqrt{2}}{2} \left\{ \ln \left| \frac{\sqrt{4x^2-4x+8}-\sqrt{7}}{2x-1} - \sqrt{7}-2\sqrt{2} \right| - \ln \left| \frac{\sqrt{4x^2-4x+8}-\sqrt{7}}{2x-1} - \sqrt{7}+2\sqrt{2} \right| \right\} + C$$

$$(35) \int \frac{dx}{x\sqrt{4x^2+1}}$$

$$\tan \theta = 2x \Rightarrow \sec^2 \theta d\theta = 2dx$$



$$\begin{aligned} \int \frac{dx}{x\sqrt{4x^2+1}} &= \int \frac{\frac{\sec^2 \theta d\theta}{2}}{\frac{\tan \theta}{2} \cdot \sec \theta} = \int \frac{\sec \theta d\theta}{\tan \theta} = \int \frac{d\theta}{\sin \theta} = \int \operatorname{cosec} \theta d\theta \\ &= \ln |\operatorname{cosec} \theta - \cot \theta| + C = \ln \left| \frac{\sqrt{4x^2+1}}{2x} - \frac{1}{2x} \right| + C \end{aligned}$$

$$(36) \int \sqrt{4x^2-4x-3} dx = \int \sqrt{(2x-1)^2-4} dx$$

$$2x-1 = 2 \sec \theta \Rightarrow 2dx = 2 \sec \theta \tan \theta d\theta \Rightarrow dx = \sec \theta \tan \theta d\theta$$

$$\int \sqrt{(2x-1)^2-4} dx = \int \sqrt{4 \sec^2 \theta - 4} \sec \theta \tan \theta d\theta = 2 \int \underbrace{\tan \theta}_{u} \underbrace{\sec \theta \tan \theta d\theta}_{dv}$$

$$= 2 \left[\tan \theta \cdot \sec \theta - \int \sec^2 \theta \cdot \sec \theta d\theta \right]$$

$$= 2 \tan \theta \sec \theta - 2 \int (\tan^2 \theta + 1) \sec \theta d\theta$$

$$= 2 \tan \theta \sec \theta - 2 \int \sec \theta \tan^2 \theta d\theta - 2 \int \sec \theta d\theta$$

$$\Rightarrow 4 \int \sec \theta \tan^2 \theta d\theta = 2 \left[\tan \theta \sec \theta - \ln |\sec \theta + \tan \theta| + C \right]$$

$$\int \sqrt{4x^2-4x-3} dx = 2 \int \sec^2 \theta \tan^2 \theta d\theta$$

$$= \tan \theta \sec \theta - \ln |\sec \theta + \tan \theta| + C$$

$$= \frac{\sqrt{4x^2-4x-3}}{2} \cdot \frac{2x-1}{2} - \ln \left| \frac{\sqrt{4x^2-4x-3} + 2x-1}{2} \right| + C$$

